

ANKIT ROY

Solution Analyst | Full Stack Expertise | Low Code Platforms | Scalable Backend Architectures

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SUMMARY

Full Stack Engineer with 3+ years experience in low-code platform development and enterprise engineering. Strong in applying system design principles to architect scalable, secure and maintainable solutions. Proficient in Java Spring Boot and React.js, with proven track record of managing the full lifecycle.

WORK EXPERIENCE

Solution Analyst (Full Stack Engineer)

September 2022 – Present

Argusoft India Ltd.

Next Generation Platform Development (Business Modeller 2.0)

- **Configuration-Driven UI Generation & Deployment:** Engineered a **runtime UI generation engine** in the greenfield low-code platform that dynamically generates React applications from configuration, connects them to backend services, builds and deploys applications automatically across **all supported operating systems** enabling **100% cross-platform compatibility** across **Linux, Windows** and **macOS**.
- **Action Based Architecture (Command Pattern):** Implemented an action-driven backend architecture using the **Command Pattern**, where every change in the configuration is converted into atomic command objects - **enabling undo/redo, environment synchronisation, audibility and installable modules**.
- **Backend Data Architecture For Entity Service:** Led a team of 2 engineers to design and implement a bidirectional **entity management system**, supporting **dynamic entity creation** reverse entity registration from existing database schemas, and automatic foreign-key and relationship resolution.
- **State Synchronisation & Reliability:** Architected a state-sync and schema reconciliation engine to maintain data consistency during **database outages, connection changes, migrations** and **manual database modifications**, ensuring platform stability.
- **Performance Optimisation:** Engineered a **custom in-memory caching** layer using Java object referencing, reducing runtime configuration lookup latency by **~50%** for configuration-heavy enterprise applications.
- **OpenAPI-Driven Backend Design:** Designed backend services compliant with **OpenAPI Specifications (OAS)**, enabling visual API configuration while reducing manual API development effort by **~60%**.
- **Developer Experience & Tooling:** Integrated Monaco Editor to provide **IDE-like experience**, including IntelliSense (TypeScript and platform-supported Java), runtime file-path resolutions for multiple apps and content diffs from previous state.
- **Modularity:** Leading a team of **4 engineers** to build **installable UI module** using **React Hook Form**, providing reusable components, validations and submissions.

Legacy Platform Optimisation & Enterprise Delivery (Business Modeller)

- **Performance Engineering:** Achieved a **~40% performance improvement** by refactoring legacy HashMap-based data handling into structured DTOs, optimising SQL execution plans, batching queries, and reducing database round trips.
- **Observability & Monitoring:** Architected a decoupled logging framework using file-system-based storage and built an interactive developer dashboard, reducing **mean-time-to-resolution (MTTR)** by **~30%**.
- **Enterprise Client Delivery:** Configured, built, deployed, and supported production-grade applications for **Boston College, Jamaica Constabulary Force (JCF)**, and **Norwegian Refugee Council (NRC)**.

Full Stack Web Developer Intern

June 2020 – September 2020

HSPM Solutions LLP.

🔗 <https://tinyurl.com/hspm-intern>

- Developed a production-ready web application "**Lions Club District 3234 D2**", improving user engagement by **~40%** and supporting day-to-day operations for **Lions Club Pune**.
- Built end-to-end web solutions for **BS Auto Accessories** and **BR Engineering Works**, helping improve online presence and increasing visibility by **~50%**.
- Assisted in backend optimisation for **Retail Addict**, contributing to a **~25% reduction** in **API response time** through query optimisation and code-level improvements.
- Gained hands-on experience in **full-stack web development**, including front-end UI development, backend API integration, testing and deployment under senior engineer guidance.

EDUCATION

Bachelor of Technology (B.Tech) - Computer Science and Engineering

2018 - 2022

Gandhi Institute For Technology (Affiliated to Biju Patnaik University of Technology, Odisha)

CGPA - 9.04

Bhubaneswar

Higher Secondary - Central Board of Secondary Education (CBSE), Science

2016 - 2018

BSF Senior Secondary School

Percentage - 66.80%

Shillong

Secondary - Indian Certificate of Secondary Education (ICSE)

2005 - 2016

St. Edmund's School

Percentage - 83.16%

Shillong

PROJECT

Pehchan E-Learning (B.Tech Final Year Project)

 <https://github.com/ankit-roy2k/pehchan.git>

- Created a **touch-free attendance system** for classrooms and office environments using facial recognition, reducing manual attendance errors by **~90%**.
 - Implemented attendance capture using **surveillance cameras** in closed spaces, eliminating the need for separate attendance-making time.
 - Phase 1: Built **single-face recognition**-based attendance system for controlled environments.
 - Phase 2: Enhanced the system with **multi-face recognition** and **liveness detection** to support **30+ users simultaneously**.
 - Used the **FaceNet deep learning model** (developed by Google researchers) for face embedding and recognition.
 - Stored attendance data locally in **CSV files** and implemented a **scheduled night upload** to the central server for record consolidation and reporting.
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Shift to Smart (Client Project)

 <https://shifttosmart.com>

- Developed a **web-based** service request portal for **Shift to Smart Services Pvt. Ltd.**, enabling users to request services for **event management, digital marketing** and **admission services** to schools and colleges.
 - Extended the platform to a full-fledged **School Management System(SMS)** supporting multiple schools, used by **2** schools.
 - Integrated modules for **student management, administration**, and **operations** improving client operational efficiency by **~30%** through a centralised admin panel.
 - Implemented the application using **PHP, HTML, CSS, JavaScript, Bootstrap** and **MySQL** for database management.
 - Gained practical exposure to **client requirements gathering, full-stack web development, database design**, and **iterative feature enhancement**.
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TECHNICAL SKILLS

Programming Languages:

Java, JavaScript (ES6+), TypeScript, Python, SQL

Frameworks & Libraries:

Spring Boot, React.js, React Hook Form, Monaco Editor

Architecture, System Designs & Design Patterns:

Monolithic Architecture, Modular Design, Low Level Design (LLD), High Level Design (HLD), RESTful API Design, Command Pattern, Factory Pattern, Builder Pattern, Strategy Pattern

Database & Data Engineering:

MySQL, PostgreSQL, JDBC, Schema Design, Query Optimisation, Multi-Database Operations

DevOps & Tools:

Git, Docker, CI/CD Pipelines, Cross-Platform Deployment (Linux, Windows, macOS)

Methodologies:

Agile Methodology, Clean Code Principles, Root Cause Analysis, Debugging and Troubleshooting

Core:

Data Structures & Algorithms (DSA), Object Oriented Programming (OOP)

CERTIFICATION

Cambridge Business English Certification (Vantage)

 <https://tinyurl.com/vantage-bec>